

Lamiaa Hassan Zain

Mechatronics and AI Engineer

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GitHub · LinkedIn · YouTube · Scholar

Summary

Mechatronics and AI Engineer (M.Sc.) with strong expertise in building, optimizing, and deploying scalable AI/ML and Generative AI components. Researcher with published IEEE researches about AI and Computer Vision for Autonomous Mobile Robots Navigation. Proficient in PyTorch, AWS cloud platform (SageMaker, Lambda, Endpoints) for Deployment. Experienced in end-to-end model integration with adjacent systems, performance optimization, automated pipelines, testing, and documentation. Udacity AI Session Lead & Mentor for AWS ML and GenAI programs and "Master of Science in AI" program. Eager to apply my technical and research background to develop core AI framework components, streamline workflows, and drive product innovation.

Work history

Session Lead & Mentor

Current

Udacity. Session Lead for the Machine Learning Engineer with AWS and Generative AI programs; Mentor for the Master of Science in Artificial Intelligence X WOOLF program.

Assistant Lecturer

Oct. 2023 – Aug. 2024

Coventry University, Administrative Capital, Egypt.

Tutor

Feb. 2023 – July 2023

University of Hertfordshire, Administrative Capital — Mechatronics track.

Teaching Assistant

2018 – Oct. 2022

Nile University, Sheikh Zayed, Egypt. Recommendation letter (1), Recommendation letter (2)

Research

- Imitation Learning for Obstacle Avoidance Using End-to-End CNN-Based Sensor Fusion — [IEEE Xplore](#)
- Real-Time Obstacle Avoidance for a Mobile Robot using CNN-Based Sensor Fusion — [IEEE Xplore](#)
- Preliminary Results (YouTube)

Education

- **M.Sc. Mechatronics** — Nile University. Thesis: "Real-Time Obstacle Avoidance for a Mobile Robot in Known and Unknown Environments Using CNN-Based Sensor Fusion."
- **Pre-Master's, Mechatronics Engineering** — Nile University, Egypt. CGPA: 3.7 (A)
- **B.Sc. Mechanical Engineering (Mechatronics)** — Benha Faculty of Engineering, Egypt, 2013–2018. Grade: Very good (76.18%)

AI Projects

- **Generative AI:** Apply-Lightweight-Fine-Tuning-to-a-foundation-model — [GitHub](#)
- **Generative AI:** Build Your Own Custom Chatbot — [GitHub](#)
- **ML DevOps:** Build an ML Pipeline for short-term rental prices in NYC — [GitHub](#)
- **ML DevOps:** Predict customer churn with clean code — [GitHub](#)
- Sentiment analysis using SageMaker XGBoost — [Code](#), [video](#)
- Predict bike sharing demand with AutoGluon, AWS — [GitHub](#)
- Image classification using AWS SageMaker — [GitHub](#)
- Build a Machine Learning workflow for Scones Unlimited on Amazon SageMaker — [GitHub](#)
- HAAR Cascade classifier based face recognition — [Vid. 01](#), [Vid. 02](#)
- Starbucks predicting offers for users using RNNs — [GitHub](#)
- Plagiarism Detector — [GitHub](#)

Mechatronics Projects

- **ROS:** Go Chase It — [GitHub](#)
- **ROS:** Build my world — [GitHub](#)
- TurtleBot2 design, manufacturing, and control: [GrabCAD \(1\)](#), [video \(SharePoint\)](#)
- Smart lock case design for smart home applications — [GrabCAD](#)
- 6 wheels Piborg mobile robot design, manufacturing, and control — [GrabCAD \(1\)](#), [Drive \(2\)](#)
- 4R Serial manipulator design, manufacturing, and control — [GrabCAD \(1\)](#), [YouTube \(2\)](#)
- Ansys Simulation for 4R serial manipulator — [Google Drive](#)

Courses & certificates

- **2022** — AWS Machine Learning Engineer Nanodegree — [credential](#)
- **2022** — Fundamentals of accelerated computing with CUDA Python from Nvidia — [credential](#),
- **2020** — AWS machine learning Nanodegree course, Online on Udacity — [credential](#)
- **2020** — AWS machine learning foundations course, Online on Udacity — [credential](#)
- **2020** — SQL for data science, UCDAVIS, Online on Coursera — [credential](#)
- **2020** — Introduction to relational database and SQL, Online on Coursera — [credential](#)
- **2020** — Machine learning, Stanford University, Online on Coursera — [credential](#)
- **2020** — Introduction to SQL, Online on Data camp — [credential](#)
- **2020** — Machine learning foundations, Washington University, Online on Coursera — [credential](#)
- **2019** — Basics of machine learning and python, Markov academy, Egypt — [credential](#)

Skills

Software-based

Python: NumPy, Numba (CPU/CUDA), Pandas, OpenCV, Pillow, PyTorch; CUDA for training & deployment.

Embedded C (Micro C, Eclipse), Linux, Git

Engineering

Mechanical design (SolidWorks); circuits (Proteus); PIC/AVR; ROS; MATLAB.

Data science

AWS SageMaker: Lambda, Step Functions, Endpoints, Batch transform, Ground Truth, S3.

SQL, PL/SQL, SQL Developer, relational DBs; ML/DL; math & statistics for ML.

Other

Presentations; communication; teamwork.

■ Academic activities

- Certified ITI trainer for the “AI-Pro” program — [Google Drive](#)
- Mentor, NASA Space Apps, Cairo — [2019](#), [2020](#), [2021](#), [2023](#), [2024](#)
- Speaker for Machine Learning in Space challenges NASA space apps, 2022 — [SharePoint](#)
- Speaker for Artificial Intelligence and Machine Learning session, IEEE MUST — [Google Drive](#)
- MOT2021 conference coordinator — [Google Drive](#)
- Judge, Undergraduate research forum, Nile University — [7th](#), [10th](#), [11th](#), [12th](#)

■ Bachelor graduation project

Four DOF Robot Arm, ROS controlled, for 2D sketching, pick & place objectives. My role: 3D model design and forward & inverse kinematics (MATLAB).

[video \(1\)](#) · [video \(2\)](#)